JULIEN DUBUC

Engineering Student - Mechatronics, Robotics, AI & Embedded Systems

Seeking a 6-month final-year internship starting in February 2027

CONTACT

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Driver's License (B) & Coastal Boating License

SKILLS

Programming & AI

- Python, C/C++, Java, Arduino IDE
- AI Tools (research, code, structuring)

Robotics & Simulation

- ROS, Gazebo, MATLAB / Simulink

Design (CAD/FEA/3D)

- SolidWorks, Onshape
- SimScale (FEA)
- PrusaSlicer (3D printing)

Computer Vision & Motion Capture

- OpenCV, Model training (YOLOv8)
- 3D Triangulation, OptiTrack, Qualisys

Embedded Systems

- Arduino, MPLAB X

Tools & Data

- GitHub, LaTeX
- Google Workspace, Microsoft Office
- Gantt, RACI, Canva, Inkscape

LANGUAGES

French - Native
English - B2 - TOEIC 920/990
German - A2
Chinese - A1

SOFT SKILLS

- Communication
- Project Management
- Teamwork
- Autonomy in R&D

INTERESTS

- Trail, Weightlifting (5 years) - discipline
- Robot Club Toulon - mechanics, AI
- Travel - London, New York, Kenya

REFERENCES

Dr. Matteo Menolotto
Senior Researcher, Tyndall National Institute
matteo.menolotto@tyndall.ie +353 83 414 9343

Lyudmyla Yushchenko
Director of Studies, SeaTech
lyudmyla.yushchenko@univ-tln.fr

EDUCATION

SeaTech Engineering School - Mechatronics and Robotics Systems - Toulon *2024 – 2027*
Robotics (terrestrial, marine, and submarine), electronics, control systems, informatics, CAD.
Creativity and Innovation minor*:* entrepreneurial project, team management, marketing.

Double Degree - Master in Intelligent Robotics and Embedded Systems *2026 – 2027*
Mechanics-robotics, control systems, artificial intelligence, embedded systems.

CPGE (Intensive Preparatory Classes) MPSI - MP - Le Havre *2022 – 2024*

Baccalaureate, Highest Honors - Le Havre *2022*

PROFESSIONAL EXPERIENCE

R&D Engineering Intern - Tyndall National Institute, Cork, Ireland *May – August 2026*
Design of a compliant robotic gripper with e-textile sensors and self-locking mechanism

- Designed and optimized 2 grippers (flexible, rack-and-pinion) autonomously using Onshape (CAD), PrusaSlicer (3D Printing), and Arduino: reduced the number of parts by 3 and assembly time by 50%.
- Conducted FEA analysis via SimScale to optimize geometry and solve buckling issues.
- Characterized and integrated e-textile sensors on fingers using Inkscape and designed the electronic setup (Wheatstone bridge, AD623, Arduino UNO) for haptic feedback.
- Established testing protocols (force, kinematics, passive holding at 0W, fatigue). Post-processed OptiTrack data (10 cameras) using MATLAB. The flexible gripper was 2.54x more energy-efficient.
- Integrated the system with a robotic arm and validated safe, smart grasping of fragile objects.

Assistant Site Manager Intern - Logeo Seine, Le Havre *June – July 2025*

- Monitored construction sites and supervised workers for social housing projects.
- Created Excel spreadsheets for financial analysis of real estate projects (construction and renovation).

Physiotherapist Assistant Intern - Maison de Santé, Caux Estuaire *February 2019*

- Worked with patients and participated in their rehabilitation using complex equipment.

Logistics Assistant Intern - Auchan, Le Havre *February 2019*

- Managed stock logistics and optimized shelf stocking in a team environment.

PROJECTS (More details on my portfolio: https://julien-dubuc.github.io/portfolio/)

Vision-based 3D Positioning System for ROV - COSMER Lab, SeaTech *2025 – 2026*

- Developed a complete Python pipeline (detection, multi-camera calibration via ArUco, 3D triangulation) to track a BlueROV in a pool without physical markers (low-cost alternative to Qualisys) in a team of 4.
- Trained a YOLOv8 model for robust detection in aquatic environments.
- Validated accuracy (1.2cm at 1.5m & 4.7cm at 3m) and delivered open-source architecture on GitHub.

Terrestrial Robotics - SeaTech *2025 – 2026*

- **Autonomous navigation of a TurtleBot** - ROS/Gazebo, state machine for obstacle avoidance using Lidar.
- **Low-level control of an autonomous mobile robot** - C, MPLAB X, rangefinders, real-time avoidance.
- **Vehicle dynamics simulator** - MATLAB App Designer, evasion trajectories.

Study of Floating Breakwaters - C4 CONSULTANTS, SeaTech *2024 – 2025*

- Led a team of 10 students to study floating and underwater breakwaters, installed sensors, utilized a wave basin, and analyzed data using MATLAB.

Hexapawn AI (Reinforcement Learning) - **TIPE - Lycée François 1er** *2023 – 2024*

- Developed a self-learning AI in Python (reward and penalty system) for the game Hexapawn: modeled the complete game tree, generating an unbeatable agent.